The region that maximizes $\iiint_E (1 - x^2 - 2y^2 - 3z^2)\,\mathrm{d}V$, and that maximum

Problem

(a) Find the region E for which the triple integral

$$\iiint_E (1 - x^2 - 2y^2 - 3z^2)\,\mathrm{d}V$$

is a maximum. (b) Use a computer algebra system to calculate the exact maximum value of the triple integral in part (a).

Notation used in this solution.

- $g(x, y, z) = 1 - x^2 - 2y^2 - 3z^2$ is the integrand.
- D is the open solid ellipsoid $x^2 + 2y^2 + 3z^2 < 1$; Z is its surface $x^2 + 2y^2 + 3z^2 = 1$; N is the exterior $x^2 + 2y^2 + 3z^2 > 1$.
- A *null set* is a set of volume 0. A surface has volume 0.
- $\mathrm{d}V$ is the volume element. In Cartesian coordinates $\mathrm{d}V = \mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$, and $\mathrm{d}V = \rho^2 \sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\varphi\,\mathrm{d}\theta$ in spherical coordinates.
- E is *admissible* if it is bounded and its boundary is a finite union of smooth surfaces. Every region in this exercise is admissible.

Part 1: the region that maximizes the integral

1. The integrand is positive exactly inside one ellipsoid

The integrand depends on the three variables only through the combination $x^2 + 2y^2 + 3z^2$. Hence

$$g(x, y, z) > 0 \iff x^2 + 2y^2 + 3z^2 < 1 \iff \frac{x^2}{1^2} + \frac{y^2}{(1/\sqrt{2})^2} + \frac{z^2}{(1/\sqrt{3})^2} < 1.$$

So the level surface $g = 0$ is the ellipsoid with semi-axes 1, $1/\sqrt{2}$ and $1/\sqrt{3}$. Inside that ellipsoid $g > 0$, on it $g = 0$, and outside it $g < 0$. At the centre $g(0, 0, 0) = 1$, which is the largest value of g.

2. Why this ellipsoid maximizes the integral

Let E be any admissible region. The three sets D, Z, N cover space, so E splits into $E \cap D$, $E \cap Z$ and $E \cap N$, and

$$\iiint_E g \, \mathrm{d}V \;=\; \iiint_{E\cap D} g \, \mathrm{d}V \;+\; \iiint_{E\cap Z} g \, \mathrm{d}V \;+\; \iiint_{E\cap N} g \, \mathrm{d}V.$$

Each piece is now easy to bound.

1. *The surface piece is zero.* Z has volume 0, and $g = 0$ on Z. So $\iiint_{E\cap Z} g \, \mathrm{d}V = 0$ whatever E is.

2. *The exterior piece is at most zero.* On N the integrand is negative, so
$$\iiint_{E\cap N} g \, \mathrm{d}V \le 0,$$
with equality only if $E \cap N$ is a null set.

3. *The interior piece is at most the integral over the whole of D.* On D the integrand is positive and $E \cap D \subseteq D$, so monotonicity of the integral gives
$$\iiint_{E\cap D} g \, \mathrm{d}V \;\le\; \iiint_D g \, \mathrm{d}V.$$

Adding the three bounds,

$$\iiint_E g \, \mathrm{d}V \;\le\; \iiint_D g \, \mathrm{d}V$$

for every admissible region E. The set $E = D$ gives equality, because then $E \cap D = D$ and the other two pieces are empty. So the integral is largest for the ellipsoid $x^2 + 2y^2 + 3z^2 < 1$.

Two points make this argument complete.

The maximizer is not unique as a set, only as a region. Equality in the bound needs $E \cap N$ to be a null set and $E \cap D = D$, and it puts no condition on $E \cap Z$. Because $g = 0$ on Z, such points may be added or removed freely. This is why the closed ellipsoid $x^2 + 2y^2 + 3z^2 \le 1$, the open ellipsoid, and the closed ellipsoid with a set of surface points deleted all give the same integral. The natural answer, and the one used below, is the closed ellipsoid

$$E^* \;=\; \big\{(x, y, z) :\; x^2 + 2y^2 + 3z^2 \le 1\big\}.$$

No other region is better. The bound above is uniform in E: it does not use any property of E except admissibility. So the value at E^* is a global maximum, not only a local one, and it is attained, not only approached.

3. The region in words and in a picture

E^* is a solid ellipsoid with centre at the origin. Its longest semi-axis is 1 along the x-axis, then $1/\sqrt{2} \approx 0.7071$ along the y-axis, then $1/\sqrt{3} \approx 0.5774$ along the z-axis. Its volume, needed below, is

$$V(E^*) \;=\; \frac{4\pi}{3} \cdot 1 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} \;=\; \frac{4\pi}{3\sqrt{6}} \;=\; \frac{2\pi\sqrt{6}}{9} \;\approx\; 1.7100664.$$

Figure 1 shows the surface and the level curves of g in the cross-section $y = 0$; the level curves shrink as g grows, and the outer curve is exactly the boundary of E^*.

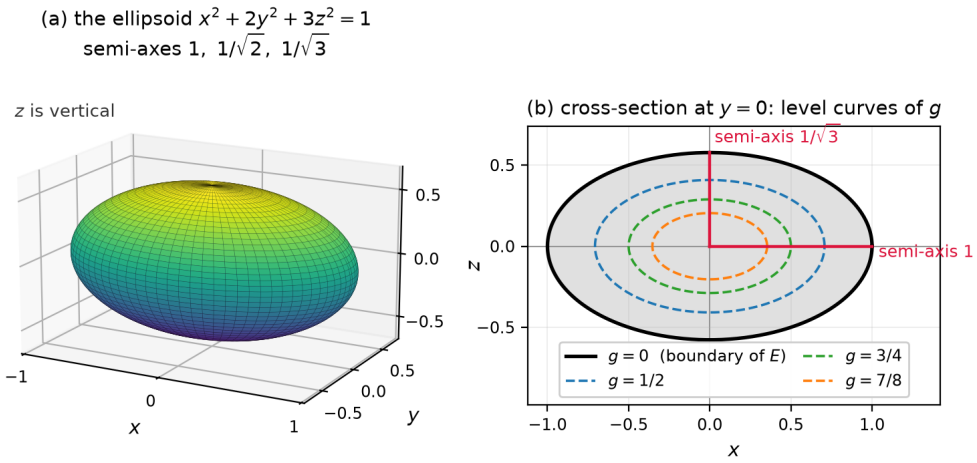


Figure 1: The region E^* where the integrand is positive. (a) The ellipsoid $x^2 + 2y^2 + 3z^2 = 1$ with semi-axes 1, $1/\sqrt{2}$, $1/\sqrt{3}$, coloured by the height z. (b) The cross-section at $y = 0$, which is the ellipse $x^2 + 3z^2 \le 1$. The nested curves are the level curves of g at $g = 1/2$, $3/4$ and $7/8$; g grows towards the centre and equals 0 on the outer curve, the boundary of E^*.

4. Numerical checks of the comparison

The scripts attached to this solution test the argument of item 2 with numbers. They use sympy for the exact work and numpy for the quadrature. Every check agrees:

- adding a cube that lies outside E^* *lowers* the integral by 6.999861;
- doubling the ellipsoid *lowers* it by 19.152744 (to -19.15);
- removing a ball of radius 0.1 from inside E^* *lowers* it by 0.004139;
- adding a ball of radius 0.1 outside E^* *lowers* it by 0.012617;
- enlarging E^* to $(1+\varepsilon)E^*$ lowers it by about $\frac{2\sqrt{6}\pi}{3}\varepsilon^2 = 5.130199\,\varepsilon^2$, which is positive for every $\varepsilon \neq 0$, so no enlargement helps.

Part 2: the exact maximum value

1. Ellipsoidal coordinates turn E^* into a ball

Put

$$u = x, \qquad v = \sqrt{2}\,y, \qquad w = \sqrt{3}\,z.$$

Then $x^2 + 2y^2 + 3z^2 = u^2 + v^2 + w^2$ and the integrand becomes $1 - u^2 - v^2 - w^2$. The inverse map is $x = u$, $y = v/\sqrt{2}$, $z = w/\sqrt{3}$, so the Jacobian is constant:

$$\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \;=\; \frac{1}{\sqrt{6}}\,\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w.$$

The ellipsoid E^* becomes the closed unit ball B in (u, v, w)-space. Therefore

$$I^* \;=\; \iiint_{E^*} g\,\mathrm{d}V \;=\; \frac{1}{\sqrt{6}} \iiint_{B} \left(1 - u^2 - v^2 - w^2\right) \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w.$$

2. The integral over the ball, in spherical coordinates

On the ball write $u = \rho\sin\varphi\cos\theta$, $v = \rho\sin\varphi\sin\theta$, $w = \rho\cos\varphi$, so $u^2 + v^2 + w^2 = \rho^2$ and the volume element is $\rho^2\sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\varphi\,\mathrm{d}\theta$. The ball is $0 \le \rho \le 1$, $0 \le \varphi \le \pi$, $0 \le \theta \le 2\pi$, and the integrand does not depend on the angles:

$$I^* = \frac{1}{\sqrt{6}}\left(\int_0^{2\pi}\mathrm{d}\theta\right)\left(\int_0^{\pi}\sin\varphi\,\mathrm{d}\varphi\right)\left(\int_0^{1}(1-\rho^2)\rho^2\,\mathrm{d}\rho\right).$$

The first integral is 2π, the second is 2, and the third is $\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$. Hence

$$I^* \;=\; \frac{1}{\sqrt{6}}\cdot 2\pi\cdot 2\cdot\frac{2}{15} \;=\; \frac{8\pi}{15\sqrt{6}} \;=\; \frac{4\pi\sqrt{6}}{45}.$$

Numerically, $I^* \approx 0.684026576086$.

3. A second exact route, by level sets

A different way to get the same number is to integrate over level sets of g. For $0 \le s \le 1$ let $E_s = \{(x,y,z) : g \ge s\}$. That is an ellipsoid similar to E^*, scaled by $\sqrt{1-s}$, so

$$V(s) \;=\; V(E_s) \;=\; \frac{4\pi}{3\sqrt{6}}(1-s)^{3/2}, \qquad V'(s) \;=\; -\frac{2\pi}{\sqrt{6}}(1-s)^{1/2}.$$

The layer formula $\iiint_E g\,\mathrm{d}V = \int_0^{\max g} V(s)\,\mathrm{d}s$ then gives

$$I^* \;=\; \int_0^1 \frac{4\pi}{3\sqrt{6}}(1-s)^{3/2}\,\mathrm{d}s \;=\; \frac{4\pi}{3\sqrt{6}}\cdot\frac{2}{5} \;=\; \frac{8\pi}{15\sqrt{6}},$$

in agreement with item 2.

4. A sanity check with the volume

With $V(E^*) = 4\pi/(3\sqrt{6})$ the mean value of the integrand on the region is

$$\frac{I^*}{V(E^*)} \;=\; \frac{8\pi}{15\sqrt{6}}\cdot\frac{3\sqrt{6}}{4\pi} \;=\; \frac{2}{5}.$$

That number is plausible: g runs from 0 on the surface to 1 at the centre, and it is larger than $1/2$ only on the small central ellipsoid with $s > 1/2$.

5. The computer algebra check over a whole family of ellipsoids

Part (b) asks for a computer algebra system. The check below does more than evaluate one integral: it shows that no ellipsoid centred at the origin does better. For the family

$$E(a,b,c) \;=\; \left\{(x,y,z) : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1\right\}$$

Ellipsoid $E(a,b,c)$	$I(a,b,c)$	I/I^*
$(1,\ 1/\sqrt{2},\ 1/\sqrt{3}) = E^*$	0.684026576086	1
best grid triple (1.0097, 0.7304, 0.5637)	0.682709696832	0.998075
(0.8, 0.6, 0.5)	0.581068977208	0.849482
(0.5, 0.5, 0.5)	0.366519142919	0.535825
(1.5, 1.5, 1.5)	−24.0331838000	−35.135

Table 1: The integral over members of the family $E(a,b,c)$, from the closed form. The claimed ellipsoid gives the largest value; enlarging the semi-axes past the optimum makes the value negative.

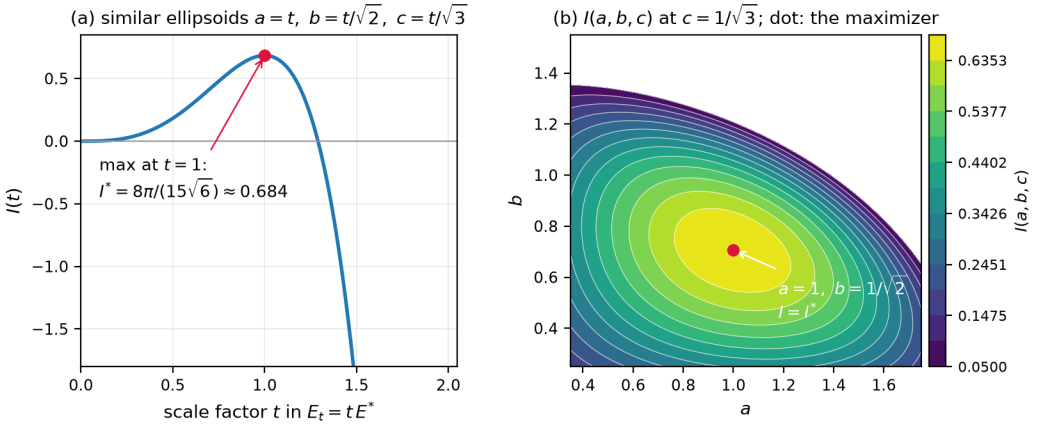


Figure 2: (a) The integral $I(t)$ for the similar family tE^*, that is $a=t$, $b=t/\sqrt{2}$, $c=t/\sqrt{3}$. The curve peaks at $t=1$ with the value $I^*=8\pi/(15\sqrt{6})$. (b) Contours of $I(a,b,c)$ at $c=1/\sqrt{3}$. The red dot marks the maximizer $a=1$, $b=1/\sqrt{2}$.

substitute $x=aX$, $y=bY$, $z=cZ$. The region becomes the unit ball, $\mathrm{d}V=abc\,\mathrm{d}V_X$, and the moments of the unit ball give $\iiint X^2\,\mathrm{d}V_X=4\pi/15$, so

$$I(a,b,c)\ =\ \frac{4\pi abc}{3}\left(1-\frac{a^2+2b^2+3c^2}{5}\right).$$

sympy confirms this formula and solves $\partial I/\partial a=\partial I/\partial b=\partial I/\partial c=0$. The only solution with $a,b,c>0$ is

$$a=1,\qquad b=\frac{1}{\sqrt{2}},\qquad c=\frac{1}{\sqrt{3}},$$

that is, E^* itself. The Hessian there has eigenvalues -1.7933, -0.7653, -0.3807, all negative, so this point is a strict local maximum. A search over 64000 triples on a $40\times40\times40$ log grid, and 2×10^6 random triples, finds no larger value: the largest ratio I/I^* in the random search is 0.999813, and every triple farther than 0.05 from the maximizer satisfies $I/I^*<0.9968$. Table 1 lists a few members, and Figure 2 shows the profile and the contour map.

6. Independent numerical confirmation

Table 2 compares the exact value with four numerical evaluations that use different methods. All of them agree with $8\pi/(15\sqrt{6})$ to the accuracy expected of the method.

Quantity	Value	Method
I^*	0.684026576086	Gauss–Legendre, ellipsoidal coordinates
I^*	0.684026576082	Gauss–Legendre, Cartesian slices
I^*	0.684291	Monte Carlo, 4×10^6 points in the bounding box
I^*	0.683839	Monte Carlo, 4×10^6 points inside E^*
$V(E^*)$	1.710066	Monte Carlo, 4×10^6 points inside E^*
mean of g on E^*	0.399961	ratio of the last two rows (2/5 exactly)

Table 2: Exact value and independent numerical values; the Monte Carlo values carry the statistical error of about 3×10^{-4} predicted by their sample size.

How to solve problems of this kind

1. **Find the sign of the integrand first.** A triple integral is largest where its integrand is positive. Write the integrand as a difference and locate its zero set; here the zero set is an ellipsoid.

2. **Split any candidate region by the sign of the integrand.** On the positive part the integral is at most the whole positive integral; on the negative part it is at most zero. Add the pieces to get a bound that holds for every region.

3. **Check whether the bound is attained.** Equality needs the region to contain the positive part (up to null sets) and to avoid the negative part. That identifies the maximizer.

4. **Discuss the zero set.** Points where the integrand is zero can be added or removed for free, so the maximizer is a region, not one exact set. State which set you report.

5. **Remove the cross terms before integrating.** A region given by $\alpha x^2 + \beta y^2 + \gamma z^2 \le 1$ becomes a ball after $u = \sqrt{\alpha}\, x$, $v = \sqrt{\beta}\, y$, $w = \sqrt{\gamma}\, z$, and the Jacobian is the constant $1/\sqrt{\alpha\beta\gamma}$. Moments of the ball, such as $\iiint X^2 \, \mathrm{d}V = 4\pi/15$, then finish the integral.

6. **Check the answer two ways.** Use an exact method (sympy, or a closed form from level sets) and an independent numerical one (Gauss–Legendre, Monte Carlo). If the two disagree, the error is in the setup, not in the arithmetic.

Answer

(a) The triple integral is largest for the solid ellipsoid

$$\boxed{E^* = \left\{(x, y, z) : \; x^2 + 2y^2 + 3z^2 \le 1\right\}}$$

which is exactly the region where the integrand $1 - x^2 - 2y^2 - 3z^2$ is positive.

(b) The maximum value of the triple integral is

$$\boxed{\iiint_{E^*} \left(1 - x^2 - 2y^2 - 3z^2\right) \mathrm{d}V \;=\; \frac{8\pi}{15\sqrt{6}} \;=\; \frac{4\pi\sqrt{6}}{45} \;\approx\; 0.6840266}$$

The two exact forms agree because $8/(15\sqrt{6}) = 8\sqrt{6}/90 = 4\sqrt{6}/45$, and $\sqrt{6} \approx 2.449490$.